Software Requirements Specification (SRS)

Project: Digital Campus Library Reservation Gateway
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Approvals

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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for the Digital Campus Library Reservation Gateway (LRG). It defines the functional, non-functional and security requirements, the external interfaces and the verification criteria for version 1.0, for use by developers, testers and evaluators. Companion documents are the Software Architecture and Design Specification (SAD) and the Software Test Plan (STP).

1.2 Scope

LRG is a web portal through which student members search the campus library catalog by ISBN, title or author, see whether a title is Available or CheckedOut, and place and manage holds on checked-out titles. The Head Librarian keeps availability and catalog data current and monitors hold demand. The system notifies the next member in a queue by email when a title is returned. Out of scope: physical check-out and check-in hardware, fines and payments, inter-library loans, and the internals of the campus SSO and mail services.

1.3 Audience

Developers, QA engineers, the Head Librarian and library IT staff, and course evaluators.

1.4 Definitions

Term	Meaning
LRG	Library Reservation Gateway (this system)
ISBN	International Standard Book Number (10 or 13 digits)
Hold	A reservation by a member on a CheckedOut title, kept in a first-come first-served queue
Ready	Hold state meaning the title has been returned and is kept for the first queued member for 72 hours
SSO	Single sign-on: the campus identity provider used for login
RBAC	Role-based access control
TLS	Transport Layer Security
API	Application Programming Interface
RTM	Requirements Traceability Matrix
WCAG	Web Content Accessibility Guidelines
UC	Use case

2. Overall description

2.1 Product perspective

LRG is a new, self-contained web application with a layered architecture (presentation, business logic, data). It depends on two external services: the campus SSO provider for authentication and an SMTP email service for notifications. It owns a relational Library Database holding titles, members' hold records and audit logs.

2.2 Major product functions

- Search the catalog by ISBN, title or author with paged results and availability status (UC-01, UC-02)
- Place, view and cancel holds; see queue position (UC-03, UC-04)
- Log in through campus SSO; guests may search only (UC-05)
- Update availability on return or checkout and promote the next hold (UC-06)
- Notify the next member by email (UC-07)
- Manage catalog entries and view a hold-demand report (UC-08, UC-09)

2.3 User roles and characteristics

Role	Characteristics	Main use cases
Guest	Not logged in; can only search	UC-01, UC-02
Student Member	Campus student; basic web skills; expects fast, clear answers; typically on a phone or laptop	UC-01 to UC-05
Head Librarian	Library staff; administers availability, catalog and demand data	UC-01, UC-05 to UC-09
Campus SSO (external)	Identity provider that authenticates users	UC-05
Email Service (external)	SMTP service that delivers notifications	UC-07

2.4 Operating environment

Server: Linux host or cloud VM running the application server and a relational database. Clients: current Chrome, Firefox and Edge, desktop and mobile. Network: campus network and internet, HTTPS only.

2.5 Constraints

- Authentication must use the campus SSO; LRG must not store passwords.
- Member data is personal data and must be protected under the institution's data-protection policy.
- Project delivered in a one-semester course timeline using open-source tools.
- Emails are sent through the institution's SMTP service; delivery delays are outside LRG's control.

2.6 Assumptions

Title availability is maintained by the Head Librarian in LRG; there is no live link to a check-out desk system in version 1.0. Every member has a campus email address.

3. External interface requirements

3.1 User interfaces

A responsive web UI with a search bar on the home page, a paginated result list, a book detail page, a My Holds page, and a librarian dashboard visible only to the Head Librarian. Accessibility: keyboard navigation, labelled form fields, sufficient colour contrast (WCAG 2.1 AA).

3.2 Hardware interfaces

None. The system runs on standard server hardware and is accessed from standard client devices.

3.3 Software interfaces

- **Campus SSO:** OpenID Connect / SAML login; returns a signed token with member id, name, email and role.
- **Email service:** SMTP over TLS for hold-ready and reminder emails.
- **Library Database:** relational database accessed only through the data layer using parameterized queries.

3.4 Communications

HTTPS with TLS 1.2 or higher for all traffic. JSON over REST between the portal front end and the application API. Failed emails are retried up to three times with back-off and then logged.

4. System features (detailed)

Each requirement lists priority, source, acceptance criteria and a test-case reference. Functional requirement IDs follow LRG-F-###.

4.1 Catalog Search (UC-01)

Description: Search the catalog by ISBN, title or author, view availability and open book details.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
LRG-F-001	The system shall allow a user to search the catalog by ISBN, title or author.	Functional	High	Student Member / Head Librarian	AC-LRG-F-001: A valid ISBN returns the matching title; title and author searches return all partial matches. Test: TC-SRCH-01, TC-SRCH-02	Library Database
LRG-F-002	The system shall validate that the search term is non-empty and properly formatted (ISBN-10/13 syntax when an ISBN is entered) before querying the catalog.	Functional	High	Student Member	AC-LRG-F-002: Empty or malformed input is rejected with a message and no database query runs. Test: TC-SRCH-03	Library Database
LRG-F-003	The system shall return results in pages showing title, author, ISBN and availability status (Available / CheckedOut).	Functional	High	Student Member	AC-LRG-F-003: Each page of 10 results shows all four fields; next and previous page controls work. Test: TC-SRCH-04	Library Database
LRG-F-004	The system shall display a book detail page and show a Place Hold option only when the title is CheckedOut.	Functional	Medium	Student Member	AC-LRG-F-004: The detail page lists full metadata; Place Hold appears only for CheckedOut titles. Test: TC-SRCH-05	Library Database
LRG-F-005	The system shall display "No results found" with advice to check spelling or broaden the search when a query matches no titles.	Functional	Low	Student Member	AC-LRG-F-005: Zero matches shows the message and no result list. Test: TC-SRCH-06	Library Database

4.2 Authentication and Access Control

Description: Campus login, guest search, role separation and session handling.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
LRG-F-010	The system shall authenticate members through campus credentials (single sign-on).	Functional	High	Student Member / Library IT	AC-LRG-F-010: Valid credentials create a session; invalid credentials show an error and create no session. Test: TC-AUTH-01, TC-AUTH-02	Campus SSO
LRG-F-011	The system shall allow guests to search the catalog without logging in and shall require login before a hold is placed.	Functional	Medium	Student Member	AC-LRG-F-011: A guest can search; selecting Place Hold redirects to login. Test: TC-AUTH-03	Campus SSO
LRG-F-012	The system shall restrict librarian functions to users holding the Head Librarian role.	Functional	High	Head Librarian / Library IT	AC-LRG-F-012: A Student Member calling any librarian function receives an access-denied response. Test: TC-AUTH-04	Campus SSO
LRG-F-013	The system shall let a user log out and shall expire a session after 30 minutes of inactivity.	Functional	Medium	Library IT	AC-LRG-F-013: After logout or 30 idle minutes, protected pages require login again. Test: TC-AUTH-05	Campus SSO

4.3 Hold and Reservation Management

Description: Place, view and cancel holds and see queue position, with per-member limits.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
LRG-F-020	The system shall allow a logged-in member to place a hold on a CheckedOut title.	Functional	High	Student Member	AC-LRG-F-020: A hold is created, appended to the title's queue and confirmed with its queue position. Test: TC-HOLD-01	Library Database, AUTH
LRG-F-021	The system shall reject a hold on an Available title and a second active hold by the same member on the same title.	Functional	High	Head Librarian	AC-LRG-F-021: Both cases return a clear rejection and create no hold record. Test: TC-HOLD-02	Library Database, AUTH

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
LRG-F-022	The system shall let a member view and cancel their active holds.	Functional	Medium	Student Member	AC-LRG-F-022: My Holds lists active holds; cancelling removes the hold and updates the queue. Test: TC-HOLD-03	Library Database, AUTH
LRG-F-023	The system shall show a member their position in a title's hold queue.	Functional	Low	Student Member	AC-LRG-F-023: The displayed position is correct and updates after earlier holds are cancelled. Test: TC-HOLD-04	Library Database, AUTH
LRG-F-024	The system shall limit each member to 5 active holds.	Functional	Medium	Head Librarian	AC-LRG-F-024: A sixth hold request is rejected with an explanatory message. Test: TC-HOLD-05	Library Database, AUTH

4.4 Notifications

Description: Email notices when a held title is ready and before a hold expires.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
LRG-F-030	The system shall notify a member by email when a title they hold becomes available.	Functional	Medium	Student Member	AC-LRG-F-030: When the first queued hold becomes Ready, an email naming the title is sent to that member. Test: TC-NOTF-01	Email service, RES
LRG-F-031	The system shall send a reminder 24 hours before a Ready hold expires.	Functional	Low	Student Member	AC-LRG-F-031: A reminder email is sent 24 hours before expiry and not after cancellation. Test: TC-NOTF-02	Email service, RES

4.5 Librarian Administration

Description: Keep availability and catalog data correct and monitor hold demand.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
LRG-F-040	The system shall let the Head Librarian update a title's availability status on return or checkout, and shall move the first queued hold to Ready on return.	Functional	High	Head Librarian	AC-LRG-F-040: Marking a title returned sets it Available and, if a queue exists, sets the first hold Ready. Test: TC-ADM-01	Library Database, AUTH
LRG-F-041	The system shall show the Head Librarian a report of titles ranked by hold-queue length.	Functional	Low	Head Librarian	AC-LRG-F-041: The report lists titles in descending queue length with counts matching the database. Test: TC-ADM-02	Library Database, AUTH
LRG-F-042	The system shall let the Head Librarian add and edit catalog entries.	Functional	Medium	Head Librarian	AC-LRG-F-042: A new or edited entry is saved and appears in search results. Test: TC-ADM-03	Library Database

5. Non-functional requirements (detailed)

NFRs are measurable and tied to tests. IDs follow LRG-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
LRG-NF-001	A catalog search shall return its first page within 2 seconds for 90% of requests with 200 concurrent users.	Performance	High	90th percentile <= 2 s in a JMeter load run. Test: TC-PERF-01
LRG-NF-002	The system shall be available 99.5% of each month during the academic term, excluding scheduled maintenance.	Reliability	High	Monitoring reports show >= 99.5% monthly uptime. Test: TC-AVL-01
LRG-NF-003	Member identity and hold history shall be visible only to the member concerned and the Head Librarian.	Security / Privacy	High	No API response contains another member's data. Test: TC-SEC-04

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
LRG-NF-004	The portal shall meet WCAG 2.1 AA and work on current Chrome, Firefox and Edge and on mobile browsers.	Usability / Accessibility	Medium	Accessibility audit passes; layout usable at 360 px width. Test: TC-UX-01
LRG-NF-005	The system shall support 5,000 registered members and 100,000 catalog titles without exceeding the LRG-NF-001 search time.	Scalability	Medium	Load test with seeded data meets LRG-NF-001. Test: TC-PERF-02
LRG-NF-006	Components shall communicate only through documented interfaces, and business-layer code shall have at least 70% unit-test coverage.	Maintainability	Medium	Coverage report >= 70%; no direct UI-to-database calls. Test: TC-MNT-01
LRG-NF-007	Hold creation, hold cancellation and librarian status changes shall be logged with user, time and action and retained for 1 year.	Audit / Data integrity	High	Every such action has a log entry retrievable for 1 year. Test: TC-AUD-01

5.1 Security

5.1.1 Security objectives

Objective	Description
SO-1 Confidentiality	Protect member identity, login data and hold history from unauthorized disclosure.
SO-2 Integrity	Ensure availability status and hold queues change only through authorized, validated and auditable actions.
SO-3 Availability	Keep search and hold services available and resistant to abuse such as brute-force login, scraping and request flooding.

5.1.2 Security requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
LRG-SR-001	TLS 1.2 or higher shall be mandatory for all connections between browser, application, SSO provider and mail server.	Security	High	A scan shows only TLS 1.2/1.3; plain HTTP redirects to HTTPS. Test: TC-SEC-01
LRG-SR-002	The system shall never store or log member passwords; authentication shall rely on campus SSO tokens.	Security	High	Database and logs contain no passwords. Test: TC-SEC-05
LRG-SR-003	Every hold and librarian operation shall be authorized on the server by role and ownership.	Security	High	Direct API calls with a Student token to librarian endpoints, or to another member's hold, return 403. Tests: TC-AUTH-04, TC-SEC-06
LRG-SR-004	All user input shall be validated and all database access shall use parameterized queries.	Security	High	SQL injection and script payloads are rejected or neutralized. Test: TC-SEC-02
LRG-SR-005	After 5 failed logins within 15 minutes, further attempts from that account or address shall be blocked for 15 minutes and the event logged.	Security	High	Sixth attempt is blocked and logged. Test: TC-AUTH-02
LRG-SR-006	Session cookies shall be marked Secure, HttpOnly and SameSite=Strict.	Security	Medium	Response headers show all three attributes. Test: TC-SEC-03
LRG-SR-007	Search and API endpoints shall be rate-limited to 60 requests per minute per client.	Security	Medium	Request 61 within a minute returns HTTP 429. Test: TC-SEC-07

6. Quality attributes and acceptance tests

- Exit criteria for acceptance: all High-priority functional requirements implemented and verified, no critical NFR or security failure, and the RTM shows every test case passed.
- Acceptance test suites: Catalog Search, Authentication, Holds, Notifications, Librarian Administration, Performance, Security, Accessibility.
- Quality attributes in priority order: security and privacy, correctness of availability and hold queues, performance, usability and accessibility, maintainability.
- Detailed test cases are listed in the Software Test Plan, Appendix A.

7. System models and diagrams

7.1 UML use-case diagrams

Figure 1 covers the member-facing use cases and Figure 2 covers librarian administration. Use cases and their primary actors:

Use case	Name	Primary actor	Related requirements
UC-01	Search ISBN Catalog	Student Member, Head Librarian, Guest	LRG-F-001 to 005
UC-02	View Book Details	Student Member, Guest	LRG-F-004
UC-03	Place Hold	Student Member	LRG-F-020, 021, 024
UC-04	View and Cancel My Holds	Student Member	LRG-F-022, 023
UC-05	Log In	Student Member, Head Librarian (Campus SSO)	LRG-F-010 to 013
UC-06	Update Availability Status	Head Librarian	LRG-F-040
UC-07	Notify Next Member	Email Service (system-triggered)	LRG-F-030, 031
UC-08	Manage Catalog Entries	Head Librarian	LRG-F-042
UC-09	View Hold Demand Report	Head Librarian	LRG-F-041

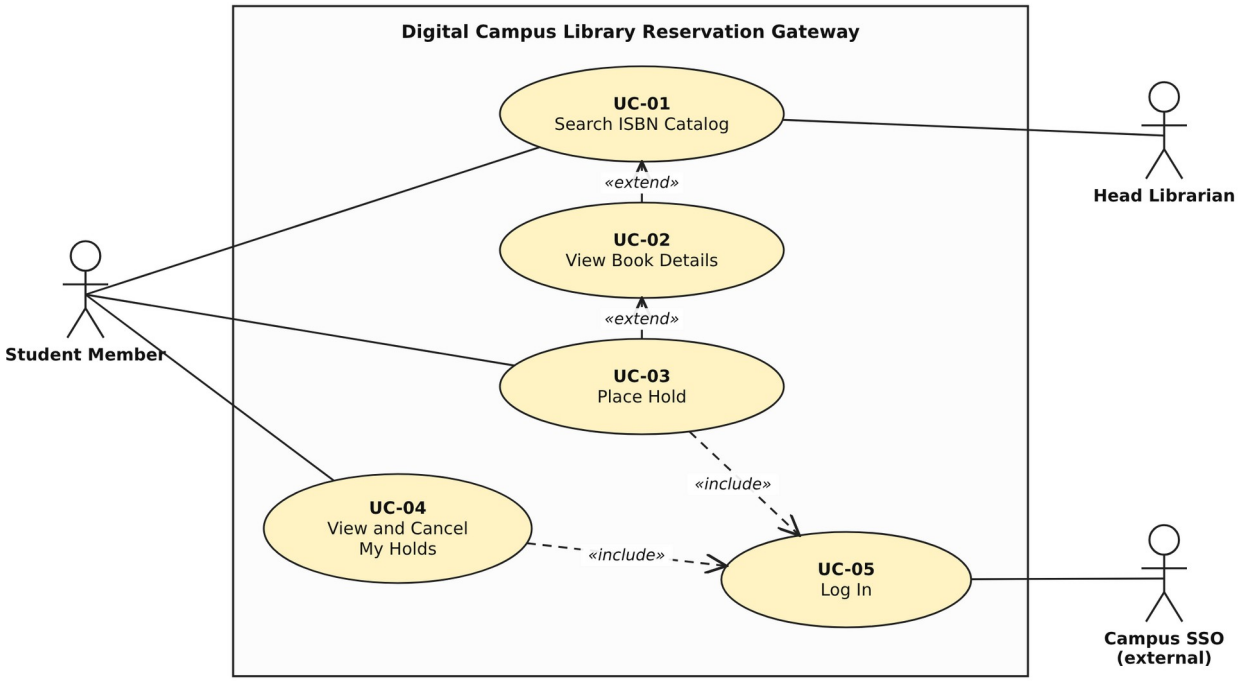


Figure 1: Use-case diagram 1 - member-facing use cases

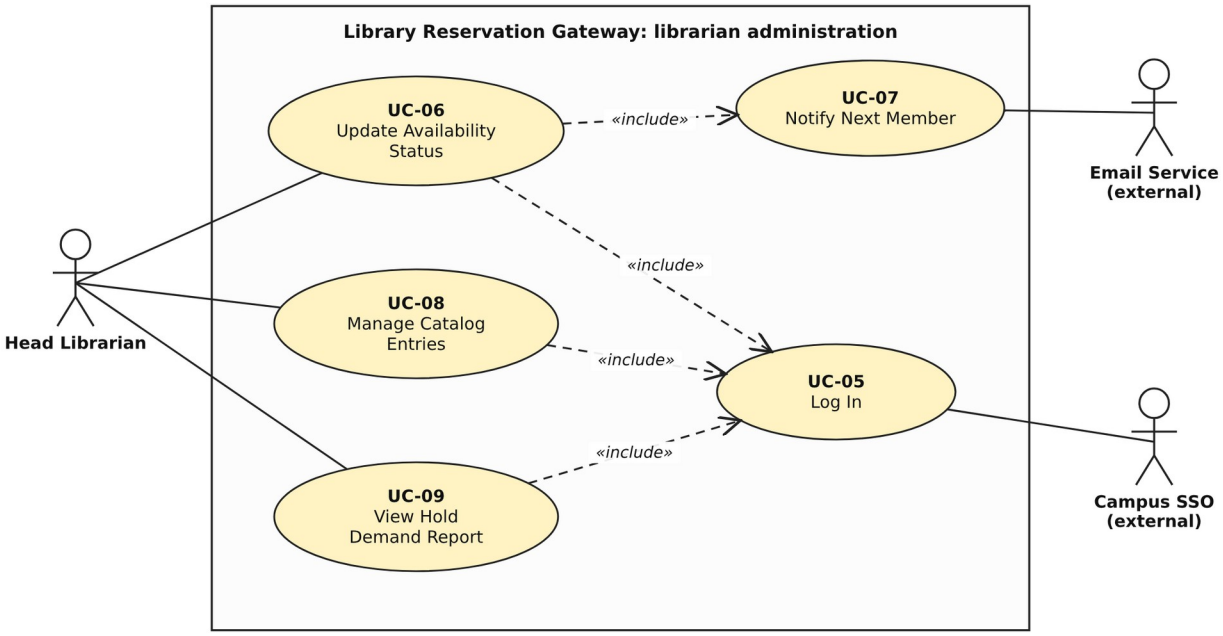


Figure 2: Use-case diagram 2 - librarian administration

7.2 UC-01 Search ISBN Catalog (specification)

Item	Description
Primary actor	Student Member (Head Librarian and Guest may also search)
Precondition	Portal is reachable; catalog contains titles
Main flow	1. Actor enters an ISBN, title or author. 2. System validates the input. 3. System queries the catalog. 4. System shows a page of results with title, author, ISBN and status. 5. Actor selects a title to see its details.
Alternate flows	A1: Input empty or malformed - show validation message, no query. A2: No matches - show "No results found" with advice.
Postcondition	Results or a message are displayed; no data is changed
Requirements	LRG-F-001 to LRG-F-005, LRG-NF-001

8. Requirements Traceability Matrix (RTM)

Status: N = Not started, P = In progress, A = Accepted. Design references point to the SAD.

Req ID	Requirement (short)	Section / Design Spec	Module	Test case(s)	Status	Comments
LRG-F-001	Search catalog	4.1 / DS-SRCH-01	Catalog Search	TC-SRCH-01, TC-SRCH-02	N	
LRG-F-002	Validate search input	4.1 / DS-SRCH-01	Catalog Search	TC-SRCH-03	N	
LRG-F-003	Paged results + status	4.1 / DS-SRCH-01	Catalog Search	TC-SRCH-04	N	
LRG-F-004	Book detail page	4.1 / DS-SRCH-01	Catalog Search	TC-SRCH-05	N	
LRG-F-005	No-results message	4.1 / DS-SRCH-01	Catalog Search	TC-SRCH-06	N	
LRG-F-010	Campus login	4.2 / DS-AUTH-01	Authentication	TC-AUTH-01, TC-AUTH-02	N	
LRG-F-011	Guest search	4.2 / DS-AUTH-01	Authentication	TC-AUTH-03	N	
LRG-F-012	Librarian role	4.2 / DS-AUTH-01	Authentication	TC-AUTH-04	N	
LRG-F-013	Logout / idle expiry	4.2 / DS-AUTH-01	Authentication	TC-AUTH-05	N	
LRG-F-020	Place hold	4.3 / DS-RES-01	Reservation Manager	TC-HOLD-01	N	
LRG-F-021	Reject invalid holds	4.3 / DS-RES-01	Reservation Manager	TC-HOLD-02	N	
LRG-F-022	View / cancel holds	4.3 / DS-RES-01	Reservation Manager	TC-HOLD-03	N	
LRG-F-023	Queue position	4.3 / DS-RES-01	Reservation Manager	TC-HOLD-04	N	
LRG-F-024	5-hold limit	4.3 / DS-RES-01	Reservation Manager	TC-HOLD-05	N	

Req ID	Requirement (short)	Section / Design Spec	Module	Test case(s)	Status	Comments
LRG-F-030	Availability email	4.4 / DS-NOT-01	Notification	TC-NOTF-01	N	
LRG-F-031	Expiry reminder	4.4 / DS-NOT-01	Notification	TC-NOTF-02	N	
LRG-F-040	Update availability	4.5 / DS-ADM-01	Reservation Manager	TC-ADM-01	N	
LRG-F-041	Hold demand report	4.5 / DS-ADM-01	Reservation Manager	TC-ADM-02	N	
LRG-F-042	Manage catalog entries	4.5 / DS-ADM-01	Catalog Search	TC-ADM-03	N	
LRG-NF-001	Search response time	5 / DS-PERF-01	SRCH / DB	TC-PERF-01	N	
LRG-NF-002	Availability 99.5%	5 / DS-DEPLOY-01	All	TC-AVL-01	N	
LRG-NF-003	Data confidentiality	5 / DS-SEC-01	RES / AUTH	TC-SEC-04	N	
LRG-NF-004	Accessibility	5 / DS-UX-01	UI	TC-UX-01	N	
LRG-NF-005	Scalability	5 / DS-PERF-01	SRCH / DB	TC-PERF-02	N	
LRG-NF-006	Maintainability	5 / DS-ARCH-01	All	TC-MNT-01	N	
LRG-NF-007	Audit logging	5 / DS-LOG-01	RES / DB	TC-AUD-01	N	
LRG-SR-001	TLS 1.2+	5.1 / DS-SEC-01	All	TC-SEC-01	N	
LRG-SR-002	No stored passwords	5.1 / DS-SEC-01	AUTH	TC-SEC-05	N	
LRG-SR-003	Server-side authorization	5.1 / DS-SEC-01	AUTH / RES	TC-AUTH-04, TC-SEC-06	N	
LRG-SR-004	Input validation / SQL	5.1 / DS-SEC-01	SRCH / RES	TC-SEC-02	N	
LRG-SR-005	Login lockout	5.1 / DS-SEC-01	AUTH	TC-AUTH-02	N	
LRG-SR-006	Cookie flags	5.1 / DS-SEC-01	AUTH / UI	TC-SEC-03	N	
LRG-SR-007	Rate limiting	5.1 / DS-SEC-01	SRCH / RES	TC-SEC-07	N	